ELEC 872 Week 6 Lab

Stuart Fong
ELEC 872 Student 2026
20287149
stuart.fong@queensu.ca

Yujong Cho
ELEC 872 Student 2026
20623720
yujong.cho@queensu.ca

1. Introduction

DeepConvLSTM [2] proposed a method for activity recognition using a machine learning model composed of CNN and LSTM layers. Their ablation is limited, so we perform a more comprehensive ablation study on the LSTM, number of convolutions, window size, regularization, learning rate warmup, and the loss function.

Our baseline model was trained according as per the workshop instructions. It follows the defaults in the original paper, which includes 4 CNN layers followed by LSTM layers with a CNN window size of 0.8s, with a dropout probability of 0.5.

2. Part A: LSTM

Our first ablation looks to see the impact of using LSTM layers compared to other choices of architecture. The first model `CNN` uses the baseline CNN detailed in the original paper [2] that replaces the LSTM layers with dense layers. The second model replaces the LSTM layers with Gated Recurrent Units (GRUs).

From the results in Table 1, we can see that the the overall performance drops with the weighted F1 dropping 2.94% and the accuracy dropping 1.53%. We see a massive drop in macro F1 of 18.70% relative to the weighted F1, likely meaning that the CNN baseline is biased towards the majority NULL class in the dataset and failing to generalize to the other classes.

For the GRU, the performance increases overall in all metrics, showing that the more simple recurrent architecture works better. This could mean that the LSTM was slightly overfitting on the training data, and so using a simpler model with fewer parameters acts as regularization for the network.

3. Part B: Convolutions

Next, we look to see the effect on the number of CNN layers for processing the features before the LSTM instead of the 4 convolution layers that are used in the baseline model. We experiment with two modes: (i) 1 convolution layer and (ii) 2 convolution layers.

Overall in the results in Table 1, we see very little change in the overall performance for both settings, with most of the metrics being within ± 1 of the reference DeepConvLSTM model performance. Though, we see that the 2 convolution layer model improves over the 1 convolution layer model. We may be able to go with a lower number of convolution layers to save on parameter count and compute resources.

4. Part C: Window Size

We try varying the window size of the CNNs for the 30Hz IMU data to see the difference from the baseline 0.8s (24 sample) window. We test the following settings: (i) 0.67s (20 sample) and (ii) 2s (60 sample).

The results in Table 1 show that both settings improve over the baseline, but the baseline 2s window is only slightly better than the reference model. The 0.67s window size shows a very high performance increase over the reference model with the weighted F1 increasing by 1.00%, the macro F1 increasing by 6.56%, and the accuracy increasing by 1.25%. This shows that smaller window sizes are likely to be more beneficial due to capturing fine-grained features than larger window sizes.

5. Part D: Regularization

For regularization, the baseline model does not show a very detailed study and simply state the dropout probability they used. We conduct more detailed experiments in the following settings: (i) setting the dropout probability to 0, (ii) adding a weight decay of 10^{-4}, (iii) Adding some noise as data augmentation for the IMU signals.

The results are shown in Table 1. Overall, we see that removing the dropout and adding data augmentation separately do not improve the performance, and are roughly equal to the baseline model. We do see an increase in model performance when using a weight decay of 10^{-4}, with the weighted F1 increasing 0.68%, the macro F1 increasing by

1.40%, and the accuracy increasing by 0.72%. This shows that regularizing the magnitude of the weights is helpful in this case for reducing overfitting in the DeepConvLSTM. We note that we previously tried a weight decay amount of 10^{-3} but the performance dropped by a significant amount, meaning that it was too strong.

6. Part E: Optimizer

The reference model uses a default constant learning rate and does not experiment with different learning rate schedules. We try to see the effect of adding a linear warmup to the training, increasing from 0 to the full 10^{-3} learning rate over the course of 5 epochs with the learning rate being updated after every step. The results are shown in Table 1. We see that the linear warmup schedule is beneficial for training the model which shows very large increase in performance overall, being the most helpful change out of all the other techniques that we tested. The weighted F1 increases by 1.11%, the macro F1 increases by 3.46, and the accuracy increases by 1.17%.

7. Part F: Class Imbalance

The original DeepConvLSTM paper does not do anything for addressing the class imbalance in these activity recognition datasets. We try using the focal loss [1] to address this. We implement the focal loss as

$$\mathcal{L}_{\text{focal}}(p_t) = -(1 - p_t)^{\gamma} \log(p_t)$$

where p_t is the probability of predicting class t. We implemented the version with no class weighing term α_t. The results are shown in Table 1. We do not see any improvement in any of the metrics, and so focal loss does not improve in this case on the cross-entropy loss used in the reference model.

8. Conclusion

From our detailed ablation study, we see that reducing the window size to 0.67s, adding weight decay of 10^{-4}, and adding a linear warmup for the first five epochs are all beneficial for improving the performance of activity recognition from IMU data. We also see that a recurrent architecture is crucial for the sequential IMU data, but that a simpler architecture like reducing the number of convolution layers and using GRU layers reduce the amount of overfitting in the model training.

Table 1. Testing results of ablation study on DeepConvLSTM. Metrics are in percentages, where higher is better. A smaller window size, weight decay, and linear warmup improve performance, while recurrent layers are crucial to the model architecture.

	Weighted F1	Macro F1	Accuracy
Part 0: Reference Model			
DeepConvLSTM	88.72	57.33	88.73
Part A: LSTM			
CNN	85.78	38.63	87.20
GRU	89.04	61.90	88.92
Part B: Convolutions			
1 conv layer	88.16	55.92	88.30
2 conv layer	89.29	59.43	89.43
Part C: Window Size			
0.67s	89.72	63.89	89.98
2s	89.39	62.45	89.14
Part D: Regularization			
no dropout	88.83	56.99	89.11
weight decay	89.40	58.73	89.45
augmentation	88.34	55.78	88.49
Part E: Optimizer			
warmup	89.83	60.79	89.90
Part F: Class Imbalance			
focal loss	88.04	52.81	87.89

References

[1] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision*, pages 2980–2988, 2017. 2

[2] Francisco Javier Ordóñez and Daniel Roggen. Deep convolutional and lstm recurrent neural networks for multimodal wearable activity recognition. *Sensors*, 16(1):115, 2016. 1